AWS Lab: CloudWatch CPU Utilization Alarm & SNS Notification Setup

Duration : 30 Min.

Scenario

An unmonitored production EC2 instance experienced CPU saturation, causing an outage that was detected only after user complaints. The operations manager mandates that an automated CloudWatch alarm must be provisioned. The alarm must monitor CPUUtilization on instance i-0a1b2c3d4e5f, fire when CPU exceeds 80% for two consecutive 5-minute periods, and publish an alert to an SNS Topic 'arn:aws:sns:us-east-1:112233445566:ops-alerts'.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon CloudWatch & Amazon SNS - **Target Instance:** `i-0a1b2c3d4e5f` - **Metric:** `CPUUtilization`, Namespace: `AWS/EC2`, Threshold: `80%` - **SNS Topic ARN:** `arn:aws:sns:us-east-1:112233445566:ops-alerts`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Formulate the AWS CLI command aws cloudwatch put-metric-alarm with all required parameters. 2. Configure MetricName: CPUUtilization, Namespace: AWS/EC2, Threshold: 80, EvaluationPeriods: 2, Period: 300. 3. Attach AlarmActions referencing the target SNS Topic ARN. 4. Save the provisioning script to ~/create_cpu_alarm.sh.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Metric configuration: put-metric-alarm uses correct CPUUtilization metric and AWS/EC2 namespace	3 Marks
TC2	Evaluation criteria: threshold set to 80 with 2 evaluation periods of 300 seconds	2 Marks
TC3	Action integration: AlarmActions correctly attaches the SNS topic ARN	3 Marks
TC4	Script validation: create_cpu_alarm.sh passes dry-run syntax verification	2 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.